

EXPLANATION OF IFR ENROUTE TERMS

FAA charts are prepared in accordance with specifications of the Interagency Air Committee (IAC), and are approved by representatives of the Federal Aviation Administration and the Department of Defense (DoD). Some information on these charts may only apply to military pilots.

The explanations of symbols used on Instrument Flight Rule (IFR) Enroute Charts and examples in this section are based primarily on the IFR Enroute Low Altitude Charts. Other IFR products use similar symbols in various colors. The chart legends portray aeronautical symbols with a brief description of what each symbol depicts. This section provides more details of the symbols and how they are used on IFR Enroute charts.

AIRPORTS

Operational airports are shown on IFR Enroute Charts.

Low Charts:

- All IAP Airports are shown on the Low Altitude Charts (US and Alaska).
- Non-IAP Airports are shown on the U.S. Low Altitude Charts (Contiguous US) have a minimum hard surface runway of 3,000'.
- Non-IAP airports are shown on the U.S. Low Altitude Alaska Charts are show if the runway is 3000' or longer, hard or soft surface.
- Public heliports with an Instrument Approach Procedure (IAP) or requested by the FAA or DoD are depicted on the IFR Enroute Low Altitude Charts.
- Seaplane bases requested by the FAA or DoD are depicted on the IFR Enroute Low Altitude Charts.

On IFR Enroute Low Altitude Charts, airport tabulation is provided that identifies airport names, IDs and the panels they are located on.

High Charts:

- Airports shown on the U.S. High Enroute Charts (Contiguous US) have a minimum hard surface runway of 5000'.
- Airports shown on the U.S. High Enroute Alaska Charts have a minimum hard surface runway of 4000'.

Charted airports are classified according to the following criteria:

LOW/HIGH ALTITUDE



Blue - Airports with an Instrument Approach Procedure and/or RADAR MINIMA published in the high altitude DoD Flight Information Publications (FLIPs)

Green - Airports that have an approved Instrument Approach Procedure and/or RADAR MINIMA published in either the U.S. Terminal Procedures Publications (TPPs) or the DoD FLIPs

Brown - Airports without a published Instrument Approach Procedure or RADAR MINIMA

Black - Foreign airports

Airports are plotted at their true geographic position.

Airports are identified by the airport name. In the case of military airports, Air Force Base (AFB), Naval Air Station (NAS), Naval Air Facility (NAF), Marine Corps Air Station (MCAS), Army Air Field (AAF), etc., the abbreviated letters appear as part of the airport name.

Airports marked "Pvt" immediately following the airport name are not for public use, but otherwise meet the criteria for charting as specified above.

Runway length is the length of the longest active runway (including displaced thresholds but excluding overruns) and is shown to the nearest 100 feet using 70 feet as the division point; e.g., a runway of 8,070' is labeled 81. The following runway compositions (materials) constitute a hard-surfaced runway: asphalt, bitumen, chip seal, concrete, and tar macadam. Runways that are not hard-surfaced have a small letter "s" following the runway length, indicating a soft surface.

AIRPORT DATA DEPICTION	
Low Altitude Minimum Operational Network Airport Designator MON CITY Airport Name (APT) (ICAO) D ★ 280 0★ 43s AWOS-3PT 128.326 (A) ★109.8 Part-time Frequency Lighting Capability: Part-time or established by NOTAM. See Chart Supplement for times of operation. Longest runway length to nearest 100 feet with 70 feet as the dividing point (add 00) s indicates soft surface ASOS/AWOS	
1. Airport elevation given in feet above or below mean sea level 2. Pvt - Private use, not available to general public 3. A solid line box enclosed the airport name indicates FAR 93 Special Requirements - see Directory/Supplement 4. "NO SVFR" above the airport name indicates FAR 91 fixed-wing special VFR flight is prohibited. 5. C or D following the airport identifier indicates Class C or Class D Airspace High Altitude - U.S. Minimum Operational Network Airport Designator MON CITY Airport Name (APT) Associated City Airport Identifier	6. Associated city names for public airports are shown above or preceding the airport name. If airport name and city name are the same, only the airport name is shown. The airport identifier in parentheses follows the airport name. City names for military and private airports are not shown. 7. Airport Ident ICAO Location Indicator shown outside contiguous U.S. 8. AFIS Alaska only High Altitude - Alaska Minimum Operational Network Airport Designator MON CITY Airport Name (APT) (ICAO) Associated City Longest runway length 00 00s 000.0 (A) ★ Part-time ATIS or AFIS Airport Elev Airport Ident. ICAO Location Indicator shown outside contiguous U.S.
LIGHTING CAPABILITY	
L Lighting Available L Pilot Controlled Lighting	★ Part-time or on request - No lighting available At private facilities- indicates no lighting information is available

A **L** symbol between the airport elevation and runway length means that runway lights are in operation sunset to sunrise. A **L** symbol indicates there is Pilot Controlled Lighting. A **L**★ symbol means the lighting is part-time or on request, the pilot should consult the Chart Supplement for light operating procedures. The Aeronautical Information Manual (AIM) thoroughly explains the types and uses of airport lighting aids.

VOR Minimum Operational Network (MON) Airports Designator

MON Airports with the **MON** Airport designator at the top of the Airport Data Block. The MON designation is to alert pilots to those airports that have retained ILS and VOR instrument approach procedures for safe recovery in the event of a GPS outage. Refer to the Aeronautical Information Manual (AIM) for expanded MON Airport guidance.

RADIO AIDS TO NAVIGATION

All IFR radio NAVAIDs that have been flight checked and are operational are shown on all IFR Enroute Charts. Very High Frequency/Ultrahigh Frequency (VHF/UHF) NAVAIDs, Very high frequency Omnidirectional Radio range (VORs), Tactical Air Navigation (TACANs) are shown in black, and Low Frequency/Medium Frequency (LF/MF) NAVAIDs, (Compass Locators and Aeronautical or Marine NDBs) are shown in brown.

On IFR Enroute Charts, information about NAVAIDs is boxed as illustrated below. To avoid duplication of data, when two or more NAVAIDs in a general area have the same name, the name is usually printed only once inside an identification box with the frequencies, TACAN channel numbers, identification letters, or Morse Code Identifications of the different NAVAIDs are shown in appropriate colors.

NAVAIDs in a shutdown status have the frequency and channel number crosshatched. Use of the NAVAID status "shutdown" is only used when a facility has been decommissioned but cannot be published as such because of pending airspace actions.

NAVIGATION AND COMMUNICATION BOXES - COMMON ELEMENTS	
LOW ENROUTE CHARTS RCO Frequencies NAVAID Name, SSV(s) FREQ, Ident, CH, Morse Code Latitude, Longitude 000.0 NAME (VL) (T) 000.0 IDT 000  N00°00.00' W000°00.00' NAME Controlling FSS Radio or FSS Name (AK)	HIGH ENROUTE CHARTS RCO Frequencies NAVAID Name Frequency, Ident, SSV(s), Channel Latitude, Longitude 000.0 NAME 000.0 IDT (H) (DL) 000 N00°00.00' W000°00.00' NAME Controlling FSS Radio or FSS Name (AK)
COMMON ELEMENTS (HIGH AND LOW CHARTS)	
RCO FREQUENCY Single Frequency Multiple Frequencies Frequencies transmit and receive except those followed by R and T: R - Receive Only T - Transmit Only	122.6 255.4 243.0 123.6 122.65 122.2 122.1R 121.5
NAVAID BOX Thin line NAVAID boxes without frequency(s) and FSS radio name indicates no FSS frequencies available. (AK only) Shadow NAVAID box indicates NAVAID and Flight Service Station (FSS) have same name.	VHF/UHF LF/MF  

FAA Chart Users' Guide - IFR Enroute Terms

NAVAID STANDARD SERVICE VOLUME (SSV) CLASSIFICATIONS

SSV Class	Altitudes	Distance (NM)
(T) Terminal	1000' to 12,000'	25
(L) Low Altitude	1000' to 18,000'	40
(H) High Altitude	1000' to 14,500' 14,500' to 18,000' 18,000' to 45,000' 45,000' to 60,000'	40 100 130 100
(VL) VOR Low	1000' to 5,000' 5,000' to 18,000'	40 70
(VH) VOR High	1000' to 5,000' 5,000' to 14,500' 14,500' to 18,000' 18,000' to 45,000' 45,000' to 60,000'	40 70 100 130 100
(DL) DME Low & (DH) DME High*	1000' to 12,900'	40 increasing to 130
(DL) DME Low	12,900' to 18,000'	130
(DH) DME High	12,900' to 45,000' 45,000' to 60,000'	130 100

* Between 1000' to 12,900', DME service volume follows a parabolic curve used by flight management computers.

Notes: For NAVAIDs with two SSVs, the SSV for each component is shown in paired parentheses with the VOR SSV shown first followed by the DME or TACAN SSV. Additionally, High Altitude facilities provide Low Altitude and Terminal service volume and Low Altitude facilities provide Terminal service volume. Altitudes are with respect to the station's site elevation. Coverage is not available in a cone of airspace directly above the facility. In some cases local conditions (terrain, buildings, trees, etc.) may require that the service volume be restricted. The public shall be informed of any such restriction by a remark in the NAVAID entry or by a NOTAM.

DISTANCE MEASURING EQUIPMENT

Facilities that operate in the "Y" mode for DME reception

(Y)

VOICE COMMUNICATIONS VIA NAVAID

Voice Transmitted

112.6

No Voice Transmitted

111.0**NAVAID SHUTDOWN STATUS****VHF/UHF****LF/MF****PART TIME OR ON-REQUEST****VHF/UHF****LF/MF****AUTOMATED WEATHER BROADCAST SERVICES**

ASOS/AWOS - Automated Surface Observing Station/Automated Weather Observing Station

VHF/UHF

LF/MF



Automated weather, when available, is broadcast on the associated NAVAID frequency.

LATITUDE AND LONGITUDE

Latitude and Longitude coordinates are provided for those NAVAIDs that make up part of a route/airway or a holding pattern. All TACAN facilities will include geographic coordinates.

LOW ENROUTE**HIGH ENROUTE**

N00°00.00' W000°00.00'

N00°00.00'
W000°00.00'

AIRSPACE INFORMATION

CONTROLLED AIRSPACE

Controlled airspace consists of those areas where some or all aircraft are subjected to air traffic control within the following airspace classifications of A, B, C, D, & E.

Air Route Traffic Control Centers (ARTCC) are established to provide Air Traffic Control to aircraft operating on IFR flight plans within controlled airspace, particularly during the enroute phase of flight. Boundaries of the ARTCCs are shown in their entirety using the symbol below.



When Controller Pilot Data Link Communication (CPDLC) exists for an ARTCC, the text CPDLC (LOGON KUSA) will be shown parallel to the boundary above or below the ARTCC identification as shown below.



The responsible ARTCC Center names are shown adjacent and parallel to the boundary line. ARTCC sector frequencies are shown in boxes outlined by the same symbol.



Class A Airspace is depicted as open area (white) on the IFR Enroute High Altitude Charts. It consists of airspace from 18,000 Mean Sea Level (MSL) to FL600.

Class B Airspace is depicted as screened blue area with a solid line encompassing the area.

Class C Airspace is depicted as screened blue area with a dashed line encompassing the area with a letter “C” enclosed in a box following the airport name.

Class B and Class C Airspace consist of controlled airspace extending upward from the surface or a designated floor to specified altitudes, within which all aircraft and pilots are subject to the operating rules and requirements specified in the Federal Aviation Regulations (UHF) 71. Class B and C Airspace are shown in abbreviated forms on IFR Enroute Low Altitude Charts. A general note adjacent to Class B airspace refers the user to the appropriate VFR Terminal Area Chart.

Class D Airspace (airports with an operating control tower) are depicted as open area (white) with a letter “D” enclosed in a box following the airport name.

Class E Airspace is depicted as open area (white) on the IFR Enroute Low Altitude Charts. It consists of airspace below FL180.

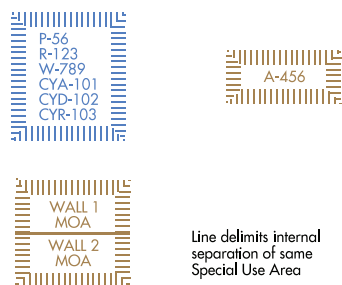
UNCONTROLLED AIRSPACE

Class G Airspace within the United States extends to 14,500’ MSL. This uncontrolled airspace is shown as screened brown.

SPECIAL USE AIRSPACE

Special Use Airspace (SUA) confines certain flight activities, restricts entry, or cautions other aircraft operating within specific boundaries. SUA areas are shown in their entirety, even when they overlap, adjoin, or when an area is designated within another area. SUA with altitudes from the surface and above are shown on the IFR Enroute Low Altitude Charts. Similarly, SUA that extends above 18,000’ MSL are shown on IFR Enroute High Altitude Charts. IFR Enroute Charts tabulations identify the type of SUA, ID, effective altitudes, times of use, controlling agency and the panel it is located on.

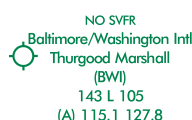
Users need to be aware that a NOTAM addressing activation will NOT be issued to announce permanently listed times of use.



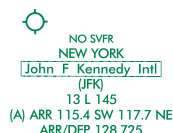
High and Low	Low Altitude Only
P - Prohibited Area	MOA - Military Operations Area
R - Restricted Area	A - Alert Area *
W - Warning Area	
* Alert Areas do not extend into Class A, B, C and D airspace, or Class E airport surface areas.	
See Airspace Tabulation on chart for complete information.	

OTHER AIRSPACE

FAR 91 Special Air Traffic Rules are shown with the type NO SVFR above the airport name.



FAR 93 Special Airspace Traffic Rules are shown with a solid line box around the airport name, indicating FAR 93 Special Requirements see Chart Supplement.



Mode C Required Airspace (from the surface to 10,000' MSL) within 30 NM radius of the primary airport(s) for which a Class B airspace is designated, is depicted on IFR Enroute Low Altitude Charts as a blue circle labeled MODE C & ADS-B OUT 30 NM.



Mode C & ADS-B Out is also required for operations within and above all Class C airspace up to 10,000' MSL, but not depicted. See FAR 91.215 and the AIM.

INSTRUMENT AIRWAYS

The FAA has established two fixed route systems for air navigation. The VOR and LF/MF system-designated from 1,200' Above Ground Level (AGL) to but not including FL 180 is shown on IFR Enroute Low Altitude Charts, and the Jet Route system designated from FL 180 to FL 450 inclusive is shown on IFR Enroute High Altitude Charts.

VOR LF/MF AIRWAY SYSTEM (IFR LOW ALTITUDE ENROUTE CHARTS)

In this system VOR airways - airways based on VOR or VORTAC NAVAIDs - are depicted in black and identified by a "V" (Victor) followed by the route number (e.g., "V12").

LF/MF airways - airways based on LF/MF NAVAIDs - are sometimes called "colored airways" because they are identified by color name and number (e.g., "Amber One", charted as "A1"). In Alaska, Green and Red airways are plotted east and west, and Amber and Blue airways are plotted north and south. Regardless of their color identifier, LF/MF airways are shown in brown.

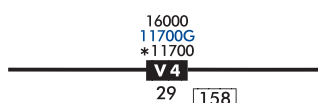
AIRWAY/ROUTE DATA

On both series of IFR Enroute Charts, airway/route data such as the airway identifications, magnetic courses bearings or radials, mileages, and altitudes (e.g., Minimum Enroute Altitudes (MEAs), Minimum Reception Altitudes (MRAs), Maximum Authorized Altitudes (MAAs), Minimum Obstacle Clearance Altitudes (MOCAs), Minimum Turning Altitudes (MTAs) and Minimum Crossing Altitudes (MCAs)) are shown aligned with the airway.

As a rule the airway/route data is charted and in the same color as the airway, with one exception. Charted in blue, Global Navigation Satellite System (GNSS) MEAs, identified with a "G" suffix, have been added to "V" and "colored airways" for aircraft flying those airways using Global Positioning System (GPS) navigation.

Airways/Routes predicated on VOR or VORTAC NAVAIDs are defined by the outbound radial from the NAVAID. Airways/Routes predicated on LF/MF NAVAIDs are defined by the inbound bearing.

- **Minimum Enroute Altitude (MEA)** - The MEA is the lowest published altitude between radio fixes that assures acceptable navigational signal coverage and meets obstacle clearance requirements between those fixes. The MEA prescribed for a Federal airway or segment, RNAV low or high route, or other direct route applies to the entire width of the airway, segment, or route between the radio fixes defining the airway, segment, or route. MEAs for routes wholly contained within controlled airspace normally provide a buffer above the floor of controlled airspace consisting of at least 300 feet within transition areas and 500 feet within control areas. MEAs are established based upon obstacle clearance over terrain and man-made objects, adequacy of navigation facility performance, and communications requirements.
- **Minimum Reception Altitude (MRA)** - MRAs are determined by FAA flight inspection traversing an entire route of flight to establish the minimum altitude the navigation signal can be received for the route and for off-course NAVAID facilities that determine a fix. When the MRA at the fix is higher than the MEA, an MRA is established for the fix and is the lowest altitude at which an intersection can be determined.
- **Maximum Authorized Altitude (MAA)** - An MAA is a published altitude representing the maximum usable altitude or flight level for an airspace structure or route segment. It is the highest altitude on a Federal airway, jet route, RNAV low or high route, or other direct route for which an MEA is designated at which adequate reception of navigation signals is assured.
- **Minimum Obstruction Clearance Altitude (MOCA)** - The MOCA is the lowest published altitude in effect between radio fixes on VOR airways, off-airway routes, or route segments that meets obstacle clearance requirements for the entire route segment and assures acceptable navigational signal coverage only within 25 statute (22 nautical) miles of a VOR. A MOCA is only shown on the Enroute Low Charts and only published when it is lower than the MEA. When shown, it is preceded by an asterisk.
- **Minimum Turning Altitude (MTA)** - Minimum turning altitude (MTA) is a charted altitude providing vertical and lateral obstruction clearance based on turn criteria over certain fixes, NAVAIDs, waypoints, and on charted route segments. When a VHF airway or route terminates at a NAVAID or fix, the primary area extends beyond that termination point. When a change of course on VHF airways and routes is necessary, the enroute obstacle clearance turning area extends the primary and secondary obstacle clearance areas to accommodate the turn radius of the aircraft. Since turns at or after fix passage may exceed airway and route boundaries, pilots are expected to adhere to airway and route protected airspace by leading turns early before a fix. The turn area provides obstacle clearance for both turn anticipation (turning prior to the fix) and flyover protection (turning after crossing the fix). Turning fixes requiring a higher MTA are charted with a flag along with accompanying text describing the MTA restriction.
- **Minimum Crossing Altitude (MCA)** - An MCA is the lowest altitude at certain fixes at which the aircraft must cross when proceeding in the direction of a higher minimum enroute IFR altitude. MCAs are established in all cases where obstacles intervene to prevent pilots from maintaining obstacle clearance during a normal climb to a higher MEA after passing a point beyond which the higher MEA applies. The same protected enroute area vertical obstacle clearance requirements for the primary and secondary areas are considered in the determination of the MCA.

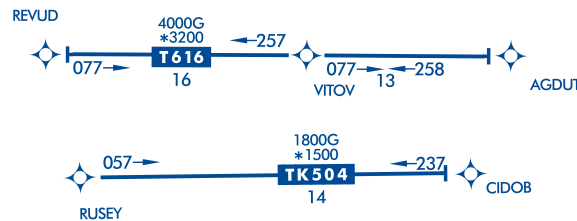


Victor Route (with RNAV/GPS MEA shown in blue)

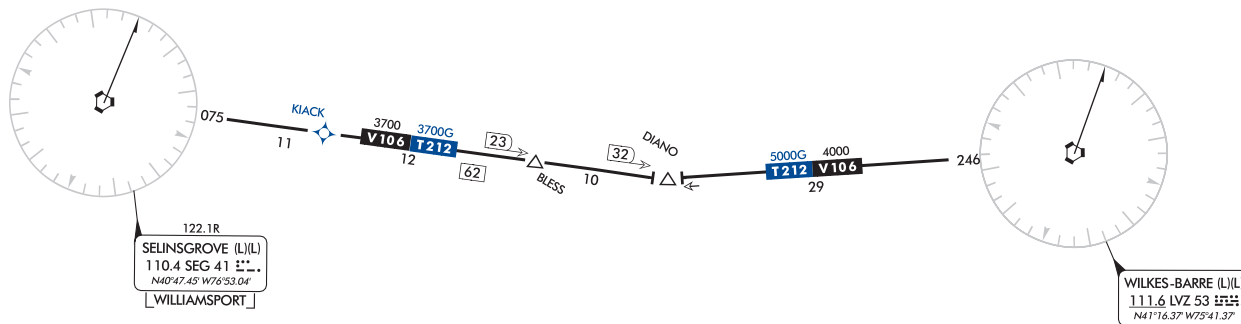
AREA NAVIGATION (RNAV) "T" ROUTE SYSTEM

The FAA has created new low altitude area navigation (RNAV) "T" routes for the enroute and terminal environments. The RNAV routes will provide more direct routing for IFR aircraft and enhance the safety and efficiency of the National Airspace System. To utilize these routes aircraft are required to be equipped with IFR approved GNSS. In Alaska, TSO-145a and 146a equipment is required.

Low altitude RNAV only routes are identified by the prefix "T", and the prefix "TK" for RNAV helicopter routes followed by a three digit number (T-200 to T-500). Routes are depicted in blue on the IFR Enroute Low Altitude Charts. RNAV route data (route line, identification boxes, mileages, waypoints, waypoint names, magnetic reference courses and MEAs) will also be printed in blue. Magnetic reference courses will be shown originating from a waypoint, fix/reporting point or NAVAID. GNSS MEA for each segment is established to ensure obstacle clearance and communications reception. GNSS MEAs are identified with a "G" suffix.

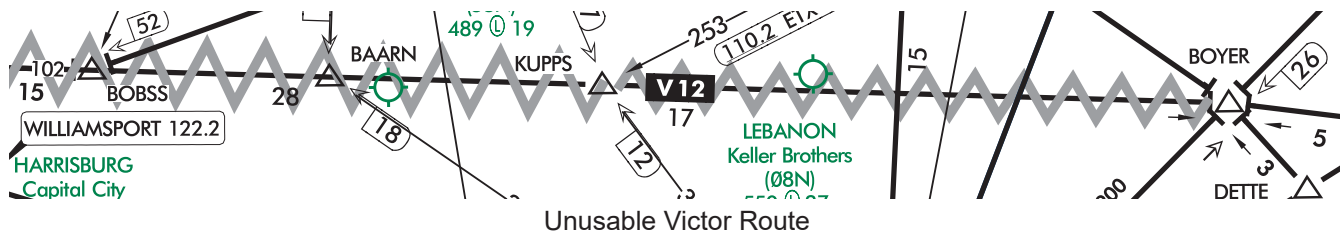


Joint Victor/RNAV routes are charted as outlined above except as noted. The joint Victor route and the RNAV route identification boxes are shown adjacent to each other. Magnetic reference courses are not shown. MEAs are charted above the appropriate identification box or stacked in pairs, GNSS and Victor. On joint routes, RNAV specific information will be printed in blue.



UNUSABLE AIRWAY/ROUTE SEGMENTS

Airway/Route segments designated by the FAA as unusable will be depicted as shown below.



Pilots should not file a flight plan for or accept a clearance that includes navigation on any route or route segment depicted as unusable. Pilots using RNAV may request ATC clearance to fly point-to-point between valid waypoints or fixes, even those on routes depicted as unusable (refer to AC 90-108 for RNAV eligibility).

Coincident Airways/Routes with Unusable Segment

When two airways/routes are coincident, but only one airway/route is designated as unusable, the following note indicating which airway the unusable symbology applies to will be placed in close proximity to the airway/route identifiers.



OFF ROUTE OBSTRUCTION CLEARANCE ALTITUDE (OROCA)

The Off Route Obstruction Clearance Altitude (OROCA) is depicted on IFR Enroute Low Altitude and Pacific charts and is represented in thousands and hundreds of feet above MSL. OROCAs are shown in every 30 x 30 minute quadrant on Area Charts, every one degree by one degree quadrant for IFR Enroute Low Altitude Charts - U.S. and every two degree by two degree quadrant on IFR Enroute Low Altitude Charts - Alaska. The OROCA is based on the highest known terrain feature or obstruction in each quadrangle, bounded by the ticked lines of latitude/longitude including data 4 NM outside the quadrant. In this example the OROCA represents 12,500 feet.

OROCA is computed just as the Maximum Elevation Figure (MEF) found on Visual Flight Rule (VFR) Charts except that it provides an additional vertical buffer of 1,000 feet in designated non-mountainous areas and a 2,000 foot vertical buffer in designated mountainous areas within the United States. Evaluating the area around the quadrant provides the chart user the same lateral clearance an airway provides should the line of intended flight follow a ticked line of latitude or longitude. OROCA altitudes are not assessed for NAVAID signal coverage, air traffic control surveillance, or communications coverage, and are published for general situational awareness, flight planning, and in-flight contingency use. OROCAs can be found over all land masses and open water areas containing man-made obstructions (such as oil rigs).

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MILITARY TRAINING ROUTES (MTRs)

Military Training Routes (MTRs) are routes established for the conduct of low-altitude, high-speed military flight training (generally below 10,000 feet MSL at airspeeds in excess of 250 knots Indicated Air Speed). These routes are depicted in brown on IFR Enroute Low Altitude Charts, and are not shown on inset charts or on IFR Enroute High Altitude Charts. IFR Enroute Low Altitude Charts depict all IFR Military Training Routes (IRs) and VFR Military Training Routes (VRs), except those VRs that are entirely at or below 1,500 feet AGL.

MTRs are identified by designators (IR-107, VR-134) that are shown in brown on the route centerline. Arrows are shown to indicate the direction of flight along the route. The width of the route determines the width of the line that is plotted on the chart:

Route segments with a width of 5 NM or less, both sides of the centerline, are shown by a .02" line.



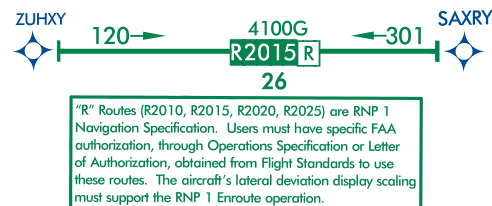
Route segments with a width greater than 5 NM, either or both sides of the centerline, are shown by a .035" line.



MTRs for particular chart pairs (ex. L1/2, etc.) are alphabetically, then numerically tabulated. The tabulation includes MTR type and unique identification and altitude range.

ALASKA CAPSTONE ROUTES (R Routes)

The Alaska Capstone program is an initiative to enhance safety and improve the efficient use of the navigable airspace in Alaska. R Routes are low altitude RNAV Routes that require specific FAA authorization. The routes are published on the L-1 Alaska Low Altitude Enroute Chart. The R Routes and associated notes are published in green with a "R" identifier followed by the route number, followed by a negative "R" identifier. An associated boxed note, also in green, is shown in close proximity to the routes.

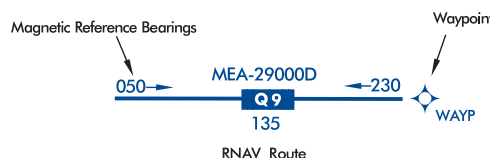


JET ROUTE SYSTEM (HIGH ALTITUDE ENROUTE CHARTS)

Jet routes are based on VOR or VORTAC NAVAIDs, and are depicted in black with a "J" identifier followed by the route number (e.g., "J12"). In Alaska, Russia and Canada some segments of jet routes are based on LF/MF NAVAIDs.

AREA NAVIGATION (RNAV) "Q" ROUTE SYSTEM (IFR ENROUTE HIGH ALTITUDE CHARTS)

The FAA has adopted certain amendments to Title 14, Code of Federal Regulations, which paved the way for the development of new area high altitude navigation (RNAV) "Q" routes in the U.S. National Airspace System (NAS). These amendments enable the FAA to take advantage of technological advancements in navigation systems such as the GPS. RNAV "Q" Route MEAs are shown when other than FL 180 MEAs for DME/DME/Inertial Reference Unit (IRU) RNAV aircraft have a "D" suffix.



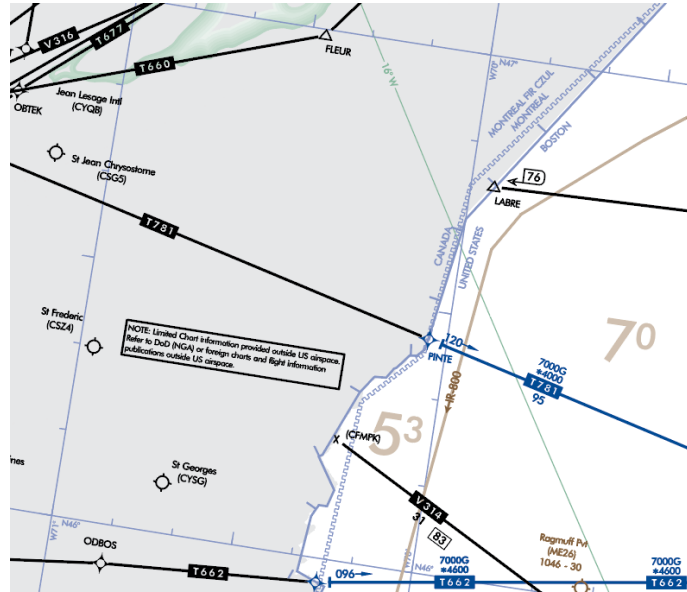
RNAV routes and associated data are charted in blue. "Q" Routes on the IFR Gulf of America charts are shown in black. Magnetic reference courses are shown originating from a waypoint, fix/reporting point, or NAVAID.

Joint Jet/RNAV route identification boxes will be located adjacent to each other with the route charted in black. With the exception of Q-Routes in the Gulf of America GNSS or DME/DME/IRU RNAV are required, unless otherwise indicated. Q-Routes in Alaska are GNSS Only. Altitude values are stacked highest to lowest.



FOREIGN AREAS

Limited data is depicted in areas outside the U.S. on all appropriate Enroute U.S., Alaska and Pacific High/Low IFR aeronautical charts. The data provided outside the U.S. is for situational awareness and transition planning. Areas outside of the U.S. have been skeletonized and symbolized in black and/or a black screen applied. This highlights the intent that U.S. charts should only be used for navigation within domestic airspace and charts of other countries should be used within their respective airspace.



TERRAIN CONTOURS ON AREA CHARTS

Based on a recommendation of the National Transportation Safety Board, terrain contours have been added to the Enroute Area Charts and are intended to increase pilots' situational awareness for safe flight over changes in terrain. The following Area Charts portray terrain: Anchorage, Denver, Fairbanks, Juneau, Los Angeles, Nome, Phoenix, San Francisco, Vancouver and Washington.

When terrain rises at least a 1,000 feet above the primary airports' elevation, terrain is charted using shades of brown with brown contour lines and values. The initial contour will be 1,000 or 2,000 feet above the airports' elevation. Subsequent intervals will be 2,000 or 3,000 foot increments.

Contours are supplemented with a representative number of spots elevations and are shown in solid black. The highest elevation on an Area Chart is shown with a larger spot and text.

The following boxed note is added to the affected Area Charts.

NOTE: TERRAIN CONTOURS HAVE BEEN
ADDED TO THOSE AREA CHARTS WHERE
THE TERRAIN ON THE CHART IS 1000 FOOT
OR GREATER THAN THE ELEVATION
OF THE PRIMARY AIRPORT

IFR ENROUTE LOW / HIGH ALTITUDE SYMBOLS (U.S., PACIFIC AND ALASKA CHARTS)

AIRPORTS

Airport Data - Low/High Altitude Civil



Seaplane - Civil



Civil And Military



Heliport



Military



Emergency Use Only



Facilities in BLUE or GREEN have an approved Instrument Approach Procedure and/or RADAR MINIMA published in either the FAA Terminal Procedures Publication or the DoD FLIPs. Those in BLUE have an Instrument Approach Procedure and/or RADAR MINIMA published at least in the High Altitude DoD FLIPs. Facilities in BROWN do not have a published Instrument Procedure or RADAR MINIMA. Facilities in BLACK are foreign airports.

All IAP Airports are shown on the Low Altitude Charts.

Non-IAP Airports shown on the U.S. Low Altitude Charts have a minimum hard surface runway of 3000'.

Airports shown on the U.S. High Altitude Charts have a minimum hard surface runway of 5000'.

Airports shown on the Alaska High Altitude Charts have a minimum hard or soft surface runway of 4000'.

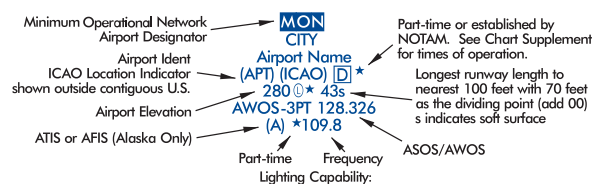
Associated city names for public airports are shown above or preceding the airport name and city name are the same only the airport name is shown. City names for military and private airports are not shown.

The airport identifier in parentheses follows the airport name or Pvt.

Pvt - Private Use

AIRPORT DATA DEPICTION

Low Altitude



1. Airport elevation given in feet above or below mean sea level

2. Pvt - Private use, not available to general public

3. A solid line box enclosed the airport name indicates FAR 93 Special Requirements - see Directory/Supplement

4. "NO SVFR" above the airport name indicates FAR 91 fixed-wing special VFR flight is prohibited.

5. [C] or [D] following the airport identifier indicates Class C or Class D Airspace

6. Associated city names for public airports are shown above or preceding the airport name. If airport name and city name are the same, only the airport name is shown. The airport identifier in parentheses follows the airport name. City names for military and private airports are not shown.

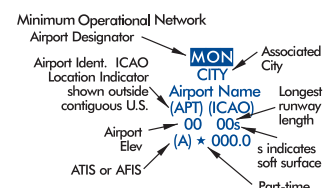
7. Airport Ident ICAO Location Indicator shown outside contiguous U.S.

8. AFIS Alaska only

High Altitude - U.S.



High Altitude - Alaska



Airports (Continued)

LIGHTING CAPABILITY

- L

Lighting Available
- ★

Part-time or on request
- Ⓛ

Pilot Controlled Lighting
- No lighting available
At private facilities- indicates no lighting information is available

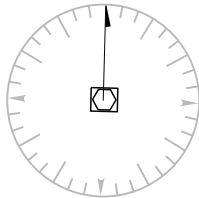
RADIO AIDS TO NAVIGATION

NAVAIDS						Reporting Function
VOR	VOR/DME	TACAN	DME	NDB	NDB/DME	
						Non Compulsory Reporting or Off Airway
						Compulsory Reporting

Note: VHF/UHF is depicted in Black. LF/MF is depicted in Brown. RNAV is depicted in Blue

Compass Roses

VHF/UHF



LF/MF



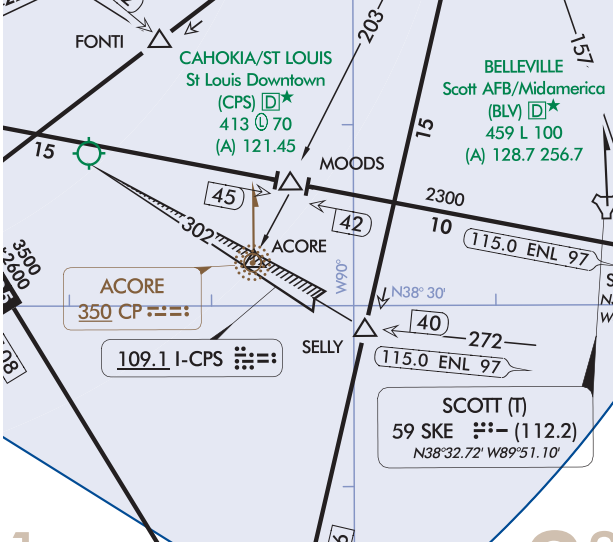
Compass Roses are oriented to Magnetic North of the NAVAID that cannot be adjusted to the charted isogonic values.

Compass Locator Beacon

LOW ALTITUDE



Chart Example : Enroute Low L-27 US



RADIO AIDS TO NAVIGATION (Continued)

LOW ALTITUDE

ILS LOCALIZER

ILS Localizer Course with additional navigation function

ILS Localizer Back Course with additional navigation function

HIGH ALTITUDE - ALASKA

VOR/DME RNAV WAYPOINT DATA

Coordinates

NOWEL

N60°29.04' W148°28.51'

115.3 MDO 297.8°-90.5

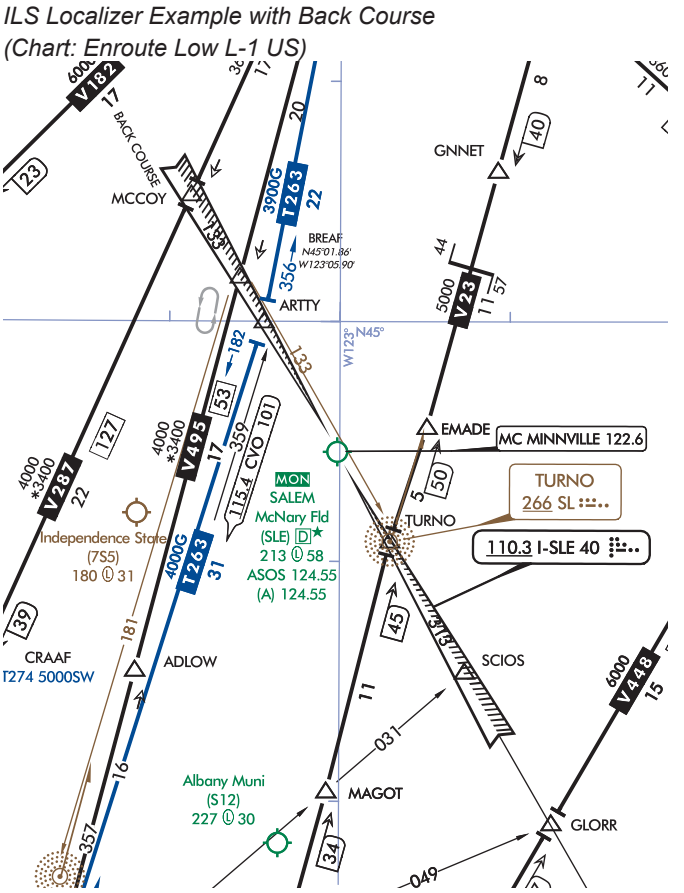
133

Frequency

Identifier

Reference Facility Elevation

Radial/Distance (Facility to Waypoint)



NAVIGATION AND COMMUNICATION BOXES - COMMON ELEMENTS

LOW ENROUTE CHARTS

RCO Frequencies

NAVAID Name, SSV(s)

FREQ, Ident, CH, Morse Code

Latitude, Longitude

000.0

NAME (VL) (T)

000.0 IDT 000

N00°00.00' W000°00.00'

Controlling FSS Radio or FSS Name (AK)

NAME

HIGH ENROUTE CHARTS

RCO Frequencies

NAVAID Name

Frequency, Ident, SSV(s), Channel

Latitude, Longitude

000.0

NAME

000.0 IDT (H) (DL) 000

N00°00.00' W000°00.00'

Controlling FSS Radio or FSS Name (AK)

NAME

COMMON ELEMENTS (HIGH AND LOW CHARTS)

RCO Frequency

Single Frequency

Multiple Frequencies

Frequencies transmit and receive except those followed by R and T:

R - Receive Only T - Transmit Only

122.6

255.4

243.0 123.6 122.65

122.2 122.1R 121.5

NAVAID Box

Thin line NAVAID boxes without frequency(s) and FSS radio name indicates no FSS frequencies available.

(AK only) Shadow NAVAID box indicates NAVAID and Flight Service Station (FSS) have same name.

VHF/UHF

LF/MF

RADIO AIDS TO NAVIGATION (Continued)

Navigation and Communication Boxes - Common Elements

NAVAID STANDARD SERVICE VOLUME (SSV) CLASSIFICATIONS

(VL), (T), etc. indicate SSV. See "NAVAID STANDARD SERVICE VOLUME (SSV) CLASSIFICATIONS" on page 66 or the Chart Supplement for SSV Altitude and Range Boundaries.

(T) (L) (VL) (H) (VH) (DL) (DH)

DISTANCE MEASURING EQUIPMENT

Facilities that operate in the "Y" mode for DME reception

(Y)

VOICE COMMUNICATIONS VIA NAVAID

Voice Transmitted

112.6

No Voice Transmitted

111.0

NAVAID SHUTDOWN STATUS

VHF/UHF

LF/MF

PART TIME OR ON-REQUEST

VHF/UHF

LF/MF

AUTOMATED WEATHER BROADCAST SERVICES

ASOS/AWOS - Automated Surface Observing Station/Automated Weather Observing Station

VHF/UHF

LF/MF

LATITUDE AND LONGITUDE

Latitude and Longitude coordinates are provided for those NAVAIDs that make up part of a route/airway or a holding pattern. All TACAN facilities will include geographic coordinates.

LOW ENROUTE

N00°00.00' W000°00.00'

HIGH ENROUTE

N00°00.00'
W000°00.00'

Navigation and Communication Boxes - Examples

LOW ENROUTE CHARTS

VOR

R - Receive only 122.1R

Controlling FSS Name - ANDERSON

122.1R
ALLENDALE (L)
116.7 AID
N33°00.72' W81°17.53'
[ANDERSON]

(T) - Service Volume

POLK (T)
108.4 FXU

Receive & Transmit on 122.35

(T) - Service Volume

Latitude and Longitude

Controlling FSS Name - MACON

122.35
TIFT MYERS (T)
112.5 IFM
N31°25.72' W83°29.33'
[MACON]

HIGH ENROUTE CHARTS

VOR

CECIL
117.9 VQQ (H)
N30°12.78'
W81°53.45'

RADIO AIDS TO NAVIGATION (Continued)

Navigation And Communication Boxes - Examples (Continued)

LOW ENROUTE CHARTS

VOR/DME

No Voice Communications
(Y) Mode DME

SAWMILL (H)(H)
113.75 SWB 84(Y)

R - Receive only 122.1R
Controlling FSS Name - BUFFALO

122.1R
ROCKDALE (L)(L)
112.6 RKA 73
N42°27.98' W75°14.36'
[BUFFALO]

(AK only) Shadow NAVAID Box
FSS Associated with NAVAID

119.1
MIRABEL (L)(L)
116.7 YMX 114
N45°53.30' W74°22.54'

TACAN

TACAN Channels are without
voice but not underlined

SANTA ROSA (L)
63 NGS
N30°36.91' W86°56.24'

Part Time NAVAID

PENSACOLA (L)
*119 NPA
N30°21.48' W87°18.99'

VORTAC

Shutdown status

255.4 243.0 122.55 121.5
ALEXANDRIA (H)(H)
116.1 AEX 108
N31°15.40' W92°30.06'
[DE RIDDER]

BRUNSWICK (L)(L)

N43°52.41' W69°55.31'

DME

DME Channel, Ident, Morse Code,
VHF Frequency

MOULTRIE (L)
25 MGR
N31°04.94' W83°48.25'

NDB

A - ASOS/AWOS Available

SILVER BAY
350 BFW

Shutdown status

SHEMYA

N52°43.32' E174°03.62'

NDB/DME

No Voice Communications
(Y) Mode DME

122.3
CAPE LISBURNE
385 LUR 20(Y) (108.35)
N68°52.28' W166°04.56'
[KOTZEBUE]

(AK only) Shadow NAVAID Box
FSS Associated with NAVAID

123.6
ILIAMNA
411 ILI 91 (114.4)
N59°44.68' W154°54.58'

HIGH ENROUTE CHARTS

VOR/DME

Off Route (Greyed NAVAID Box
and NAVAID)

ITHACA
111.8 ITH (L)(D) 55

DME in Y Mode

ELMIRA
109.65 ULW (L)(L) 33(Y)
N42°05.65'
W77°01.49'

(AK only) Shadow NAVAID Box
FSS Associated with NAVAID

119.1
MIRABEL
116.7 YMX (L)(L) 114
N45°53.30'
W74°22.54'

TACAN

Off Route

TYNDALL
64 PAM (H) (133.7)
N30°04.44'
W85°34.34'

Off Route - Part Time NAVAID
(Greyed NAVAID Box and NAVAID)

PENSACOLA
119 NPA (L) (117.2)
N30°21.48'
W87°18.99'

VORTAC

Off Route (Greyed NAVAID Box
and NAVAID)

122.55
ALEXANDRIA
116.1 AEX (H)(H) 108
N31°15.40'
W92°30.06'
[DE RIDDER]

HANDLE
114.3 HLL (L)(D) 90

DME

DME Channel, Ident,
VHF Frequency

DUNKIRK
109 DKK (H) (116.2)

NDB

FORT DAVIS
529 FDV
N64°29.68'
W165°18.91'

NDB/DME

No Voice Communications
(Y) Mode DME

CAPE NEWENHAM
385 EHM 18(Y) (108.15)
N58°39.38'
W162°04.42'

(AK only) Shadow NAVAID Box
FSS Associated with NAVAID

ILIAMNA
411 ILI 91 (114.4)

Notes: Morse Code is not shown on High NAVAID Boxes.

RADIO AIDS TO NAVIGATION (Continued)

Stand Alone Flight Services and Communication Outlets

Flight Service Station (FSS) - Alaska Only

Shadow NAVAID boxes indicate Flight Service Station (FSS) locations. Frequency 122.2 (Conterminous U.S) and 121.5 and 243.0 (Alaska) are available at many FSSs and are not shown. All other frequencies are shown above the box.

Certain FSSs provide Local Airport Advisory (LAA) on 123.6.

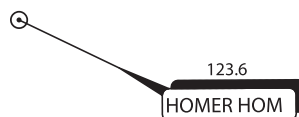
Frequencies transmit and receive except those followed by R and T:

R - Receive Only

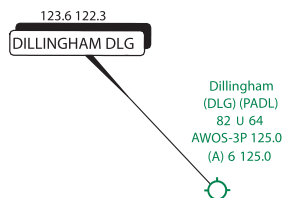
T - Transmit Only

LOW CHARTS

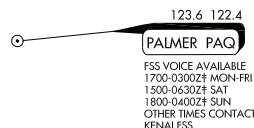
Stand Alone FSS



Stand Alone FSS Associated with an Airport



Part-time FSS

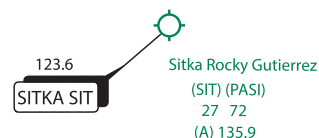


HIGH CHARTS

Stand Alone FSS



Stand Alone FSS Associated with an Airport



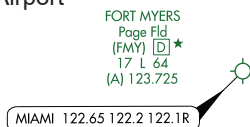
Remote Communications Outlet (RCO)

Thin line NAVAID boxes without frequencies and controlling FSS name indicate no FSS frequencies available. Frequencies positioned above the thin line boxes are remotized to the NAVAID sites. Other frequencies at the controlling FSS named are available, however altitude and terrain may determine their reception.

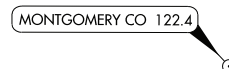
Stand Alone RCO



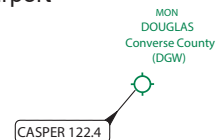
RCO Associated/Co-located with an Airport



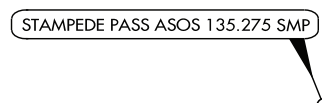
Stand Alone RCO



RCO Associated/Co-located with an Airport



Stand Alone AWOS & ASOS



AIRSPACE INFORMATION

Airway/Route Types
Low and High Enroute Airway Data:

Low Enroute Charts

Victor Airways

V0

LF/MF Airway

A0

RNAV T Route

T000

GNSS Required

RNAV TK Helicopter Route

TK000

GNSS Required

Preferred Direction Victor Route

V0

Unusable Route Segment

Military Training Routes (MTR)

MTRs 5NM or less both sides of centerline

IR-000

VR-000

MTRs greater than 5NM either or both sides of centerline

IR-000

VR-000

Arrow indicates direction of route

See MTR tabulation for altitude range information

All IR and VR MTRs are shown except those VRs at or below 1500' AGL

CAUTION: Inset charts do not depict MTRs

Alaska Capstone Routes (R Routes)

R0000R

"R" Routes (R2010, R2015, R2020, R2025) are RNP 1 Navigation Specification. Users must have specific FAA authorization, through Operations Specification or Letter of Authorization, obtained from Flight Standards to use these routes. The aircraft's lateral deviation display scaling must support the RNP 1 Enroute operation.

Low and High Enroute Charts

ATS Route

A0

A0

Oceanic Route

A00

A00

VHF/UHF Data is depicted in Black.

LF/MF Data is depicted in Brown.

RNAV Route data is depicted in Blue

High Enroute Charts

Jet Routes

J000

Atlantic Routes

AR0

AR0

Bahama Routes

BROL

BROL

RNAV Q Routes

Q00

Alaska Q Routes require GNSS and radar surveillance. Within the CONUS, GNSS or DME/DME/IRU RNAV required, unless otherwise indicated. DME/DME/IRU aircraft require radar surveillance.

Preferred Direction Jet Routes

J0

Preferred Direction RNAV Q Routes

Q0

Preferred Direction ATS Route

R000

Unusable Route Segment

Substitute Route

All relative and supporting data shown in brown.

See NOTAMS or appropriate publication for specific information.

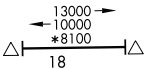
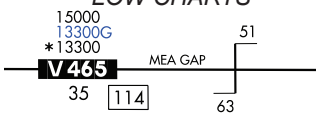
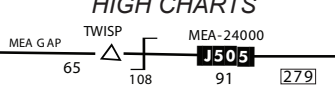
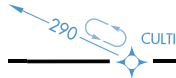
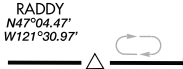
FAA Chart Users' Guide - IFR Enroute Symbolry

81

Airspace Information (Continued)

FIXES		REPORTING FUNCTION	WAYPOINTS
VHF/UHF	LF/MF	Compulsory Position Reporting	RNAV
		Non-Compulsory Position Reporting	
N25°46.47' W76°16.28'	N29°36.00' W88°01.00'	Fix or Waypoint Coordinates <i>Fix Coordinates are shown for compulsory, offshore and holding fixes.</i> <i>Waypoints Coordinates are shown when waypoint is not part of a RNAV route and when located on or beyond the boundary of the U.S. Continental Control (12 mile limit).</i>	N44°25.36' W64°11.00'
		Off-set arrows indicate facility forming a fix - Arrow points away from the VHF/UHF NAVAID - Arrow points towards the LF/MF NAVAID	N/A
		Distance Measuring Equipment (DME) Fix <i>Denotes DME fix (distance same as airway / route mileage)</i>	N/A
VHF/UHF		Distance Measuring Equipment (DME) Fix <i>Denotes DME fix (encircled mileage shown when not otherwise obvious)</i>	RNAV N/A
		Example: <i>First segment, 5NM; second segment 10NM; total mileage provided in encircled DME arrow.</i>	N/A
		Total Mileages between Compulsory Reporting Points or NAVAIDs <i>Note: All mileages are in Nautical Miles</i>	RNAV N/A
229	149		N/A
54	125	MILEAGE BETWEEN OTHER FIXES, NAVAIDS AND/OR MILEAGE BREAKDOWN	125
		Mileage Breakdown or Computer Navigation Fix (CNF) <i>Five letter identifier in parentheses indicates CNF with no ATC function</i>	N/A
		FACILITY LOCATOR BOATS <i>Crosshatch indicates Shutdown status of NAVAID</i>	N/A
		RADIAL OUTBOUND FROM A VHF/UHF NAVAID <i>All Radials are magnetic.</i>	N/A
	N/A	BEARING INBOUND TO AN LF/MF NAVAID <i>All Bearings are magnetic.</i>	N/A
N/A			
N/A	N/A	MAGNETIC REFERENCE BEARING , outbound from a NAVAID or Fix <i>Note: Not shown on joint Victor/RNAV or Jet/RNAV Routes.</i>	000 →

Airspace Information (Continued)

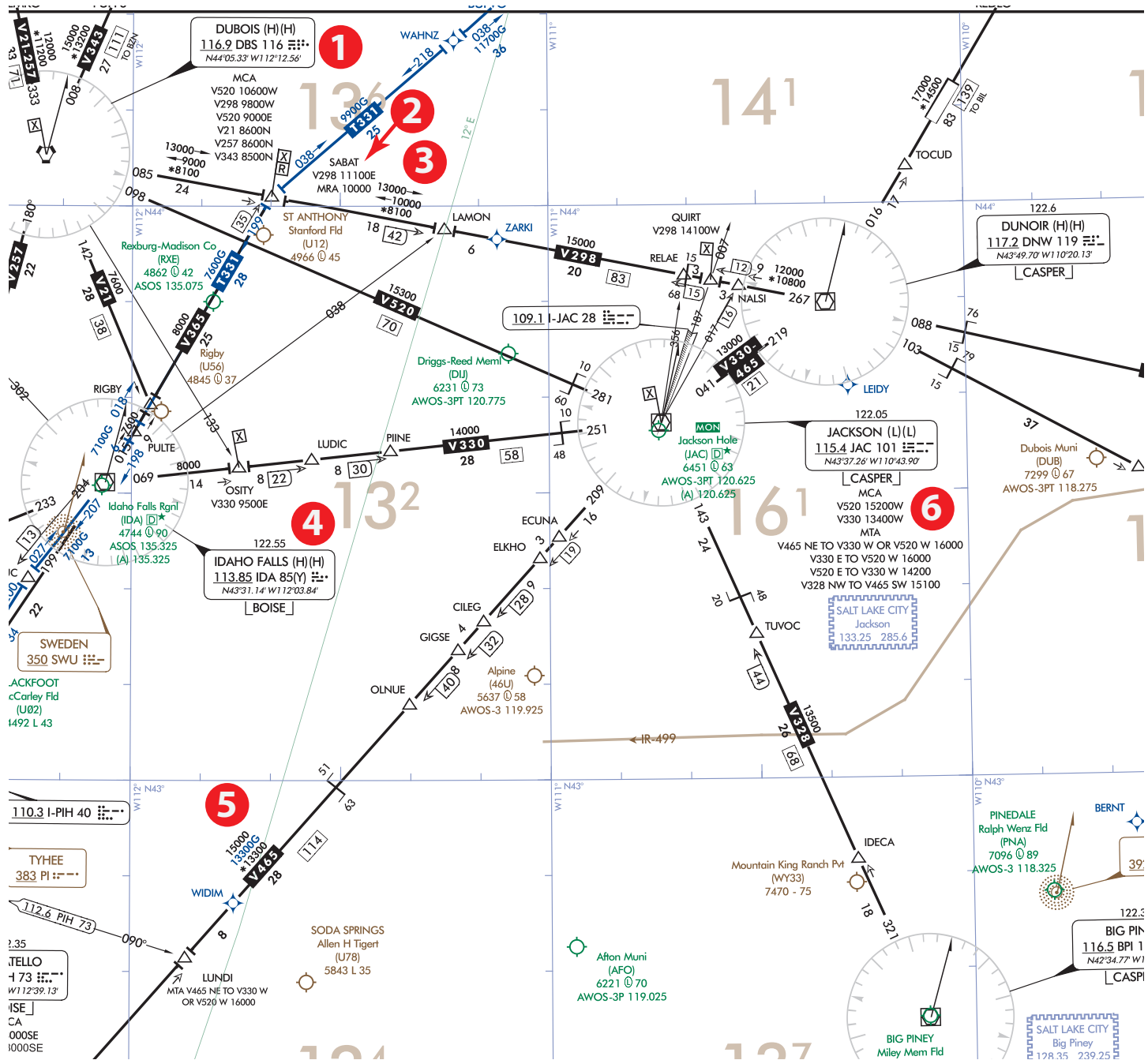
VHF/UHF LOW CHARTS	LF/MF LOW CHARTS	MINIMUM ENROUTE ALTITUDE (MEA) All Altitudes Are MSL Unless Otherwise Noted.	RNAV LOW CHARTS
0000	0000	Directional MEAs	0000G
		MEAs are shown on IFR High Altitude Charts when MEA is other than 18,000'.	HIGH CHARTS
HIGH CHARTS	HIGH CHARTS		MEA for GNSS RNAV aircraft MEA-24000G
MEA-29000	MEA-FL240		MEA for DME/DME/IRU RNAV aircraft MEA-24000D
LOW CHARTS  HIGH CHARTS 		MINIMUM ENROUTE ALTITUDE (MEA) GAP MEA is established when there is a gap in navigation signal coverage.	N/A
LOW / HIGH CHARTS	LOW / HIGH CHARTS	Maximum Authorized Altitude (MAA) All Altitudes Are MSL Unless Otherwise Noted.	LOW / HIGH CHARTS
MAA-00000	MAA-00000	MAAs are shown on IFR High Altitude Charts when MAA is other than 45,000'.	MAA-00000
LOW CHARTS	LOW CHARTS	Minimum Obstruction Clearance Altitude (MOCA) All Altitudes Are MSL Unless Otherwise Noted.	LOW CHARTS
*0000	*0000		*0000
LOW CHARTS	LOW CHARTS	Minimum Turning Altitude (MTA) and Minimum Crossing Altitude (MCA) See Low Enroute Chart Example below for examples of both MTAs and MCAs.	LOW CHARTS
			
		MINIMUM RECEPTION ALTITUDE (MRA)	N/A
		ALTITUDE CHANGE MEA, MOCA and/or MAA change at other than NAVAIDs	
LOW / HIGH CHARTS	LOW / HIGH CHARTS	CHANGEOVER POINT Changeover Point giving mileage to NAVAIDs (Not shown at midpoint locations.)	N/A
			
HOLDING PATTERNS RNAV Holding Pattern Magnetic Reference Bearing is determined by the isogonic value at the waypoint or fix. Holding Pattern with maximum restriction airspeed 210K applies to altitudes 6000' to and including 14000'. 175K applied to all altitudes. Airspeed depicted is Indicated Airspeed (IAS)			
			
			

AIRSPACE INFORMATION (Continued)

Enroute Chart Examples

Low Enroute Chart

FAA Chart Users' Guide - IFR Enroute Symbolism

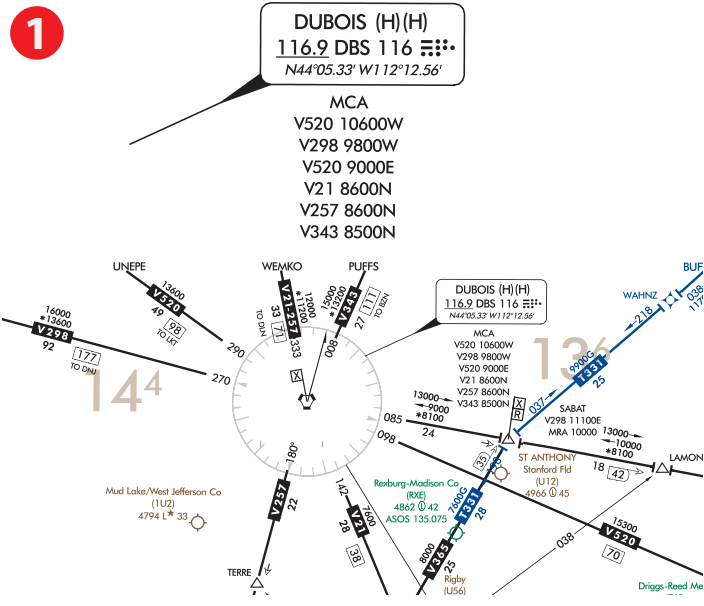


AIRSPACE INFORMATION (Continued)

Enroute Chart Examples
Low Enroute Chart (Continued)

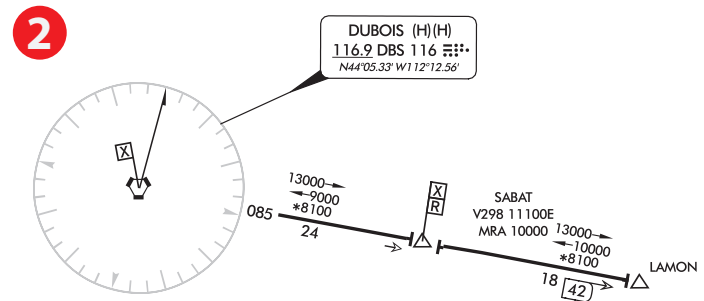
Reference Number

Description



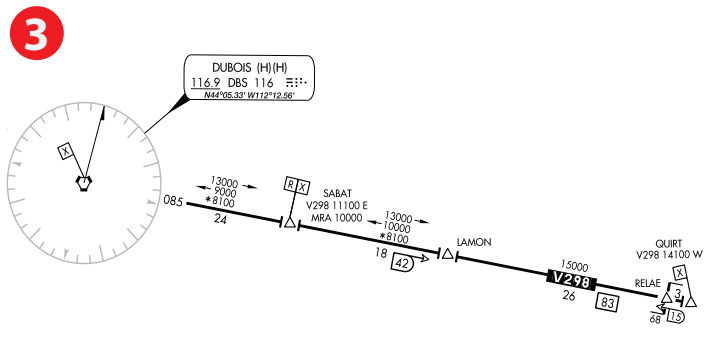
Multiple MCAs at a NAVAID

V21 and V257 - MCA at DBS of 8600' traveling North
V298 - MCA at DBS of 9800' traveling West
V343 - MCA at DBS of 8500' traveling North
V520 - MCA at DBS of 9000' traveling East
V520 - MCA at DBS of 10600' traveling West



MCA and MRA at a Fix

MCA at SABAT on V298 of 11,100 traveling East.
MRA at SABAT of 10000.

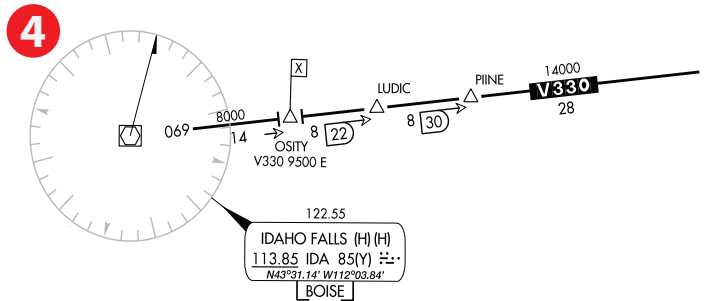


Example of MOCA and directional MEAs along a Victor Route

Traveling East from DBS, MEA 13,000' the first two segments, 15,000 along third segment.

Traveling West from QUIRT, MEA of 15,000' the first segment, MEA of 10,000 the second segment and MEA of 9,000 the third segment.

MOCA for DBS to SABAT and SABAT to LAMON segments of 8100



MCA Example

MCA at OSITY on V330. MCA of 9500' traveling East on V330 from Idaho Falls (IDA) VOR-DME.

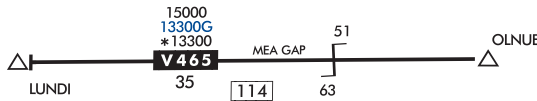
AIRSPACE INFORMATION (Continued)

Enroute Chart Examples

Low Enroute Chart (Continued)

Reference Number

5



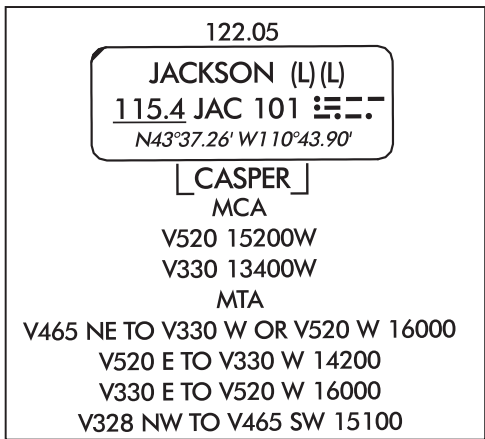
Description

MEA VHF and RNAV Example

MEA for aircraft utilizing VHF NAVAID of 15000'
MEA for aircraft utilizing RNAV of 13300'

MOCA of 13300'

6



MCA and MTA Example at a NAVAID

MCA for aircraft traveling West along V520 to cross JAC at 15200'
MCA for aircraft traveling West along V330 to cross JAC at 13400'

MTA for aircraft crossing over and turning at JAC:

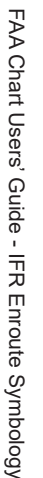
Aircraft traveling NE on V465 and turning to V330 on a W heading or turning to V520 on a W heading must turn at altitude of 16000' or higher

Aircraft traveling E on V520 and turning to V330 on a W heading must turn at altitude of 14200'

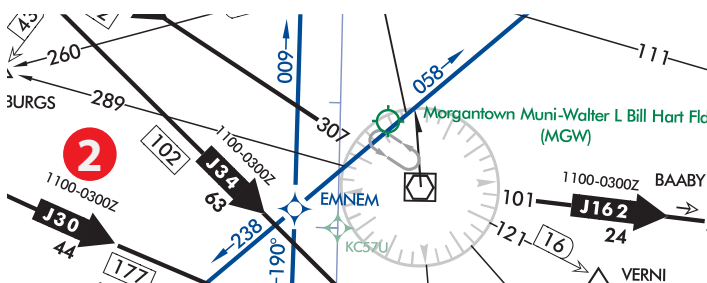
Aircraft traveling E on V330 and turning to V520 on a W heading must turn at altitude of 16000' or higher

Aircraft traveling NW on V328 and turning to V465 on a SW heading must turn at altitude of 15100' or higher.

High Enroute Chart



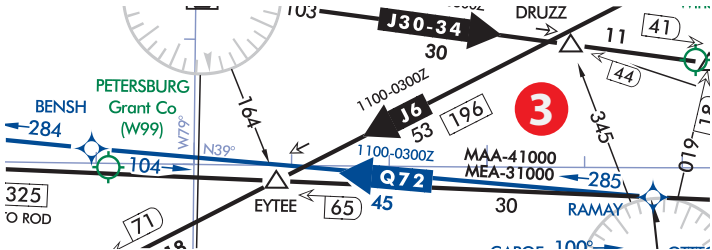
Jet Route 34 available between 1100 - 0300Z



AIRSPACE INFORMATION (Continued)

Enroute Chart Examples
High Enroute Chart (Continued)

Reference Number



Description

Preferred Directional Jet Route with Time Restrictions, MAA and MEA

Jet Route 149 available between 1100 - 0300Z
MAA - 41,000'
MEA - 31,000'

AIRSPACE BOUNDARIES

Air Defense Identification Zone (ADIZ)

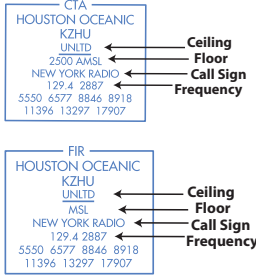
LOW / HIGH CHARTS



Adjoining ADIZ

Air Traffic Service Identification Data

LOW / HIGH CHARTS



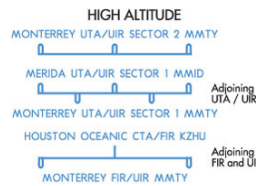
Flight Information Regions (FIR)

LOW / HIGH CHARTS



Upper Information Regions (UIR)

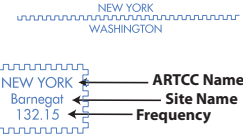
Upper Control Areas (UTA)



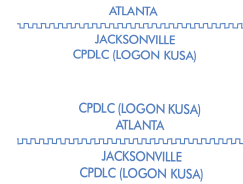
Air Route Traffic Control Center (ARTCC)

LOW / HIGH CHARTS

ARTCC Remoted Sites with discrete VHF and UHF frequencies



Air Route Traffic Control Center (ARTCC) with Controller Pilot Data Link Communications (CPDLC)



Altimeter Setting Change



Control Areas (CTA)

LOW / HIGH CHARTS



Adjoining CTA

Additional Control Areas

LOW ALTITUDE



HIGH ALTITUDE



Foreign Airspace Note

NOTE: Limited chart information provided outside U.S. airspace. Refer to DoD (NGA) or foreign charts and flight information publications outside U.S. airspace.

AIRSPACE INFORMATION (Continued)

Airspace - U.S.

Class A Open Area (White)

High Chart Only That airspace from 18,000' MSL to and including FL 600, including the airspace overlying the waters within 12 NM of the coast of the contiguous United States and Alaska and designated offshore areas, excluding Santa Barbara Island, Farallon Island, the airspace south of latitude 25° 04'00" N, the Alaska peninsula west of longitude 160°00'00" W, and the airspace less than 1,500' AGL.

Controlled Airspace

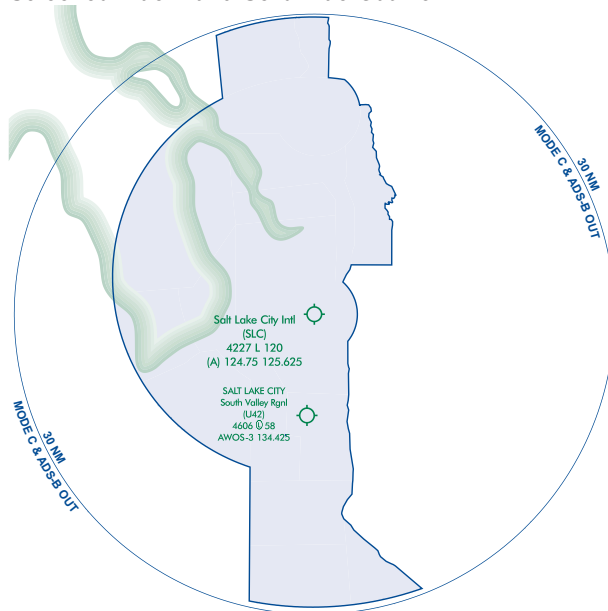
That airspace from 18,000' MSL to and including FL 450, including Santa Barbara Island, Farallon Island, the Alaska peninsula west of longitude 160°00'00" W, and designated offshore areas.

Class B

Low Chart Only

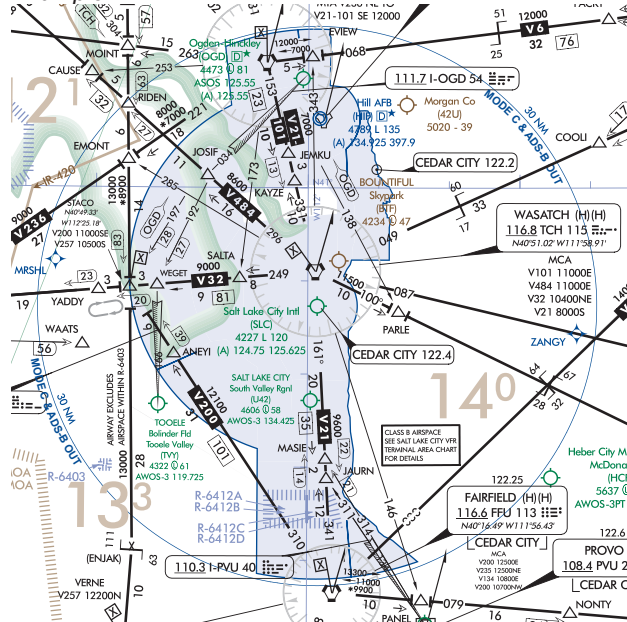
Controlled Airspace

Screened Blue with a Solid Blue Outline



That airspace from the surface to 10,000' MSL (unless otherwise designated) surrounding the nation's busiest airports. Each Class B airspace area is individually tailored and consists of a surface area and two or more layers.

Example:



FAA Chart Users' Guide - IFR Enroute Symbolology

Mode C Area

A Solid Blue Outline

Low Chart Only

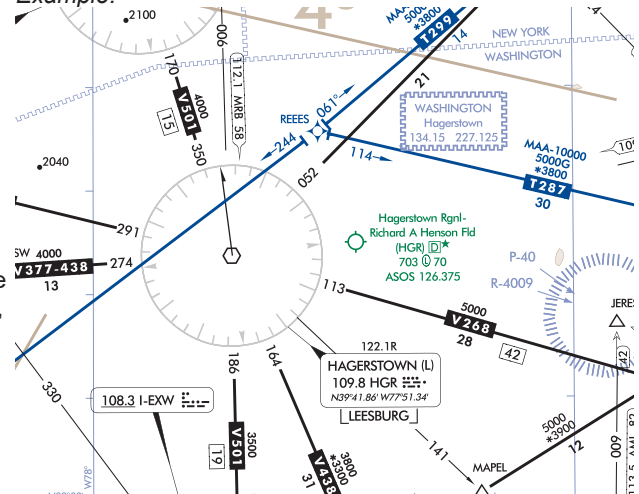
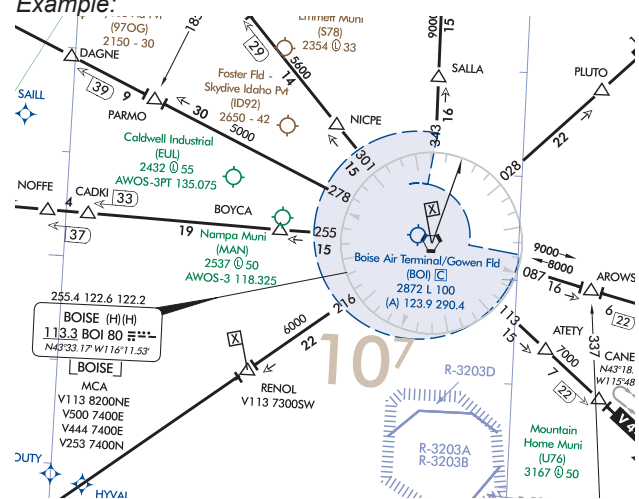
Controlled Airspace

That airspace within 30 NM of the primary airports of Class B airspace and within 10 NM of designated airports. Mode-C transponder and ADS-B Out equipment is required. (See FAR 91.215)

Example:

See Chart example above.

FAA Chart Users' Guide - IFR Enroute Symbology



Airspace Information (Continued)

AIRSPACE - U.S.

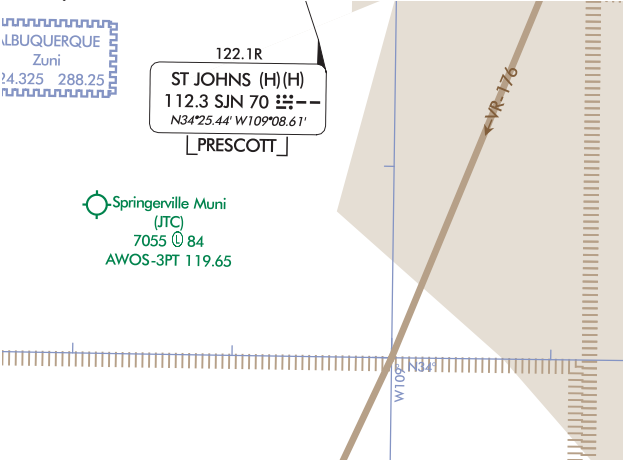
CLASS E

- Open Area (White)
- Low Chart Only
 - The controlled airspace below 14,500' MSL that is not Class B, C or D.
- Controlled Air-space
 - Federal Airways from 1,200' AGL to but not including 18,000' MSL (unless otherwise specified).
 - Other designated control areas below 14,500' MSL.
 - Not Charted
 - That airspace from 14,500' MSL to but not including 18,000' MSL, including the airspace overflying the waters within 12 NM of the coast of the contiguous United States and Alaska and designated offshore areas, excluding the Alaska peninsula west of longitude 160°00'00" W, and the airspace less than 1,500' AGL.

CLASS G

- Screened Brown Area
- High and Low Chart
 - Uncontrolled Airspace
- Low Altitude
 - That portion of the airspace below 14,500' MSL that has not been designated as Class B, C, D or E Airspace.
- High Altitude
 - That portion of the airspace from 18,000' MSL and above that has not been designated as Class A airspace.

Example:

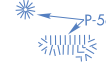


AIRSPACE INFORMATION (Continued)

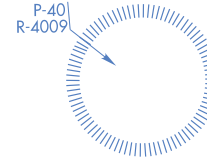
Special Use Airspace - U.S.

Low and High Charts P - Prohibited Area

Example: P-56 -
Washington DC, Area A-1 Chart



Example: P-40 and R-4009 -
Washington DC, Area A-1 Chart



R - Restricted Area

Example: R3601A -

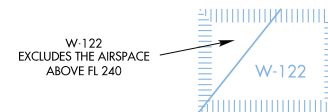
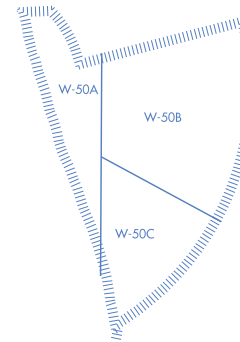


W - Warning Area

Example: W-50

See Airspace Tabulation on each chart for complete documentation information on:

Area Identification
Effective Altitude
Operating Times
Controlling Agency Voice Call



Low Charts Only

A - Alert Area

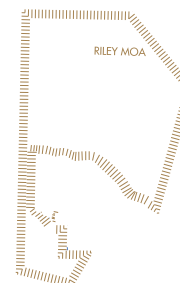
* Alert Areas do not extend into Class A, B, C and D airspace, or Class E airport surface areas.



MOA - Military Operations Area

See Airspace Tabulation on each chart for complete documentation information on:

Area Identification
Effective Altitude
Operating Times
Controlling Agency Voice Call



AIRSPACE INFORMATION (Continued)

Off Route Obstruction Clearance Altitude (OROCA)

Low Charts Only OROCA is computed similarly to the Maximum Elevation Figure (MEF) found on Visual charts except that it provides an additional vertical buffer of 1,000 feet in designated non-mountainous areas and a 2,000 foot vertical buffer in designated mountainous areas within the United States.

Example: 12,500 feet

12⁵

Example: Low L-13 Chart

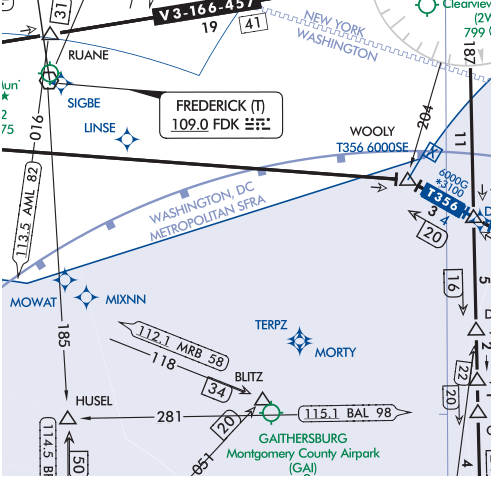


Special Flight Rules Area (SFRA)

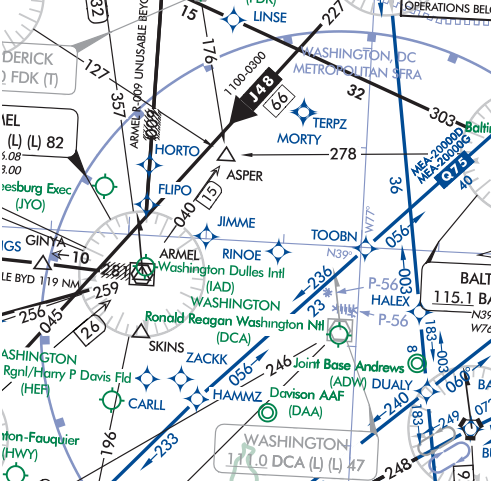
Low and High Charts SFRA Symbology



Example: Low Chart (Washington Area Chart)



Example: High Chart (H-12)



FAA Chart Users' Guide - IFR Enroute Symbology

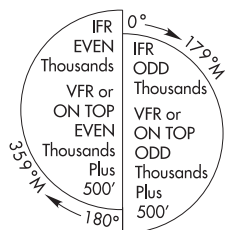
NAVIGATIONAL AND PROCEDURAL INFORMATION

Cruising Altitudes - Low Charts - U.S. Only

IFR outside controlled airspace.

IFR within controlled airspace as assigned by ATC.

ALL courses are magnetic.



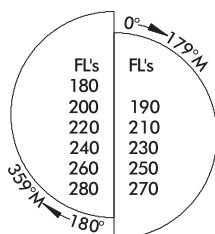
VFR above 3000' AGL
unless otherwise
authorized by ATC.

Cruising Altitudes - High Charts - U.S. Only

IFR within controlled airspace
as assigned by ATC

All courses are magnetic.

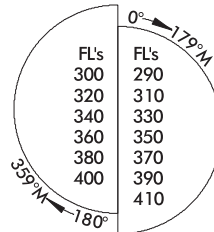
18,000' MSL to FL280



VFR or VFR On Top
add 500'

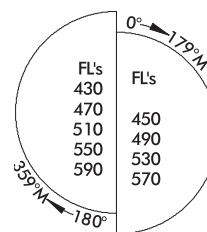
No VFR flights within
Class A Airspace above
3000' AGL unless
otherwise authorized

RVSM Levels FL290 to FL410



No VFR or VFR On Top
authorized above FL285 in
RVSM airspace.

FL430 and above



Navigational and Procedural Information (Continued)

ISOGONIC LINE AND VALUE

LOW/HIGH CHARTS

TIME ZONE

All time is Coordinated Universal Time (UTC)

Mountain Std
+7 = UTC
Central Std
+6 = UTC

During periods of Day-
lights Savings Time (DT),
effective hours will be one
hour earlier than shown.
All states observe DT ex-
cept Arizona and Hawaii

ENLARGEMENT AREA



MATCH MARK

LOW/HIGH CHARTS



MORSE CODE

A -	F	K ---	P	U ...	1 .----	6
B	G ---	L	Q ----	V	2 .----	7 .----
C ----	H	M --	R ...	W ---	3 .----	8 .----
D ---	I ..	N --	S ...	X ----	4 .----	9 .----
E .	J .----	O ---	T -	Y ----	5	0 .----
				Z ----		

CULTURE

Boundaries

International

LOW/HIGH ALTITUDE

(Not shown when coincident with ARTCC or FIR)

U.S./Russia Maritime Line

LOW/HIGH ALTITUDE



Date Line

LOW/HIGH ALTITUDE

INTERNATIONAL DATELINE
SUNDAY
MONDAY

HYDROGRAPHY

SHORELINES



TOPOGRAPHY

TERRAIN
Area Charts

